

KOLPASHEVO, Russian Federation

WMO: 292310

Lat: **58.3231N**Lon: **82.9279E**Elev: **246**StdP: **14.57**Time Zone: **7.00 (E07)**Period: **94-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-39.4	-33.3	-43.5	0.4	-38.8	-38.0	0.6	-32.9	13.1	9.0	11.8	10.2	1.8	300	0.605

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	18.0	83.2	66.4	80.2	65.2	77.0	63.6	69.4	78.5	67.4	76.3	65.5	73.8	4.9	150

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
66.3	97.6	74.6	64.2	90.6	71.2	62.4	85.0	69.7	33.8	78.6	32.1	76.6	30.6	73.8	79.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%		2.5%	5%	Mean		Standard Deviation	n=5 years		n=10 years		n=20 years		n=50 years		DB	WB
Min	Max			Min	Max		Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)		(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)		
(4) 13.1	11.4	9.9		-44.7	88.7	6.6	3.0	-49.4	90.8	-53.3	92.5	-57.0	94.2	-61.8	96.3	(4)
(5)				-42.9	73.6	6.2	3.3	-47.3	76.0	-50.9	77.9	-54.4	79.8	-58.9	82.2	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	31.8	-4.6	3.2	19.2	33.5	47.6	60.8	64.7	58.8	47.4	33.8	13.7	1.6	(6)
(7)		DBStd	26.41	16.69	14.47	11.93	10.84	10.45	8.34	6.21	6.58	8.03	9.38	15.59	16.65	(7)
(8)		HDD50	7896	1694	1312	956	499	175	15	0	9	143	506	1088	1499	(8)
(9)		HDD65	12269	2159	1732	1421	944	547	173	83	211	529	968	1538	1964	(9)
(10)		CDD50	1249	0	0	0	5	101	338	456	281	65	3	0	0	(10)
(11)		CDD65	146	0	0	0	0	7	46	74	18	1	0	0	0	(11)
(12)		CDH74	1373	0	0	0	0	99	471	638	155	10	0	0	0	(12)
(13)		CDH80	274	0	0	0	0	20	103	129	22	0	0	0	0	(13)
(14)	Wind	WSAvg	4.5	4.1	4.5	5.0	5.1	5.3	4.1	3.5	3.7	4.1	4.8	4.8	4.6	(14)
(15)	Precipitation	PrecAvg	20.2	0.9	0.7	0.8	1.0	2.0	2.3	2.8	3.0	2.0	1.8	1.5	1.3	(15)
(16)		PrecMax	28.9	2.6	1.5	2.0	2.2	5.1	4.5	8.0	7.0	4.2	3.8	3.0	2.9	(16)
(17)		PrecMin	14.2	0.1	0.1	0.1	0.2	0.6	0.3	0.0	0.8	0.4	0.7	0.4	0.3	(17)
(18)		PrecStd	3.5	0.4	0.3	0.4	0.5	1.0	1.1	1.8	1.4	0.8	0.7	0.5	0.5	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	30.8	34.1	46.8	65.2	82.6	86.9	87.2	83.2	75.4	60.9	41.3	33.2	(19)
(20)			MCWB	29.3	32.1	37.6	50.6	60.9	66.8	70.4	67.9	61.2	50.8	38.7	32.2	(20)
(21)		2%	DB	26.6	30.5	41.4	59.6	76.2	82.9	83.5	78.4	69.8	55.2	38.0	30.8	(21)
(22)			MCWB	25.3	28.5	34.3	47.7	57.8	65.1	67.9	65.4	58.5	47.0	36.0	29.6	(22)
(23)		5%	DB	22.7	27.0	38.0	54.2	70.8	79.7	80.6	74.7	65.1	50.8	35.0	27.3	(23)
(24)			MCWB	21.4	25.3	32.8	44.1	55.6	64.2	66.6	63.5	56.0	45.7	33.4	26.2	(24)
(25)		10%	DB	18.9	22.9	34.9	49.2	65.1	75.9	77.5	71.2	60.4	46.6	32.6	23.7	(25)
(26)			MCWB	17.9	21.3	31.1	40.9	52.7	62.9	65.1	62.0	53.6	42.4	31.4	22.6	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	29.2	32.1	38.3	52.1	62.7	71.1	73.1	70.3	63.1	51.8	39.5	32.7	(27)
(28)			MCD B	30.3	33.6	45.0	63.1	78.2	80.5	82.9	78.7	72.4	56.3	40.5	33.2	(28)
(29)		2%	WB	25.5	28.6	35.4	48.3	59.5	67.9	70.4	67.4	59.4	48.6	36.2	29.7	(29)
(30)			MCD B	26.7	30.2	39.7	57.3	72.2	78.5	80.0	75.5	67.7	53.9	37.7	30.7	(30)
(31)		5%	WB	21.6	25.4	33.4	44.9	56.9	65.7	68.4	65.1	57.0	45.8	33.6	26.4	(31)
(32)			MCD B	22.7	27.0	37.3	53.0	68.8	75.5	77.2	71.8	63.4	50.3	34.7	27.4	(32)
(33)		10%	WB	17.9	21.3	31.4	41.5	53.9	63.9	66.6	63.0	54.3	42.7	31.6	22.6	(33)
(34)			MCD B	18.9	22.9	34.9	48.5	63.7	73.7	75.0	69.4	59.8	46.1	32.7	23.7	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	13.9	15.8	18.0	17.8	18.4	18.5	18.0	16.8	15.1	10.9	11.4	13.1	(35)
(36)			MCDBR	12.6	11.8	18.0	24.3	26.3	25.5	22.8	23.1	23.8	17.6	9.6	13.3	(36)
(37)			MCWBR	12.3	10.8	13.0	14.4	12.1	10.9	10.1	11.8	13.3	11.1	8.9	13.1	(37)
(38)		5% WB	MCDBR	13.0	11.4	15.4	23.2	23.8	21.5	20.5	19.3	21.3	16.4	9.6	12.8	(38)
(39)			MCWBR	12.7	10.7	11.9	14.4	12.3	10.6	10.5	10.9	12.7	11.1	9.3	12.7	(39)
(40)	Clear-Sky Solar Irradiance	taub		0.278	0.312	0.359	0.384	0.396	0.403	0.442	0.409	0.364	0.337	0.296	0.278	(40)
(41)		taud		1.991	1.994	1.946	2.082	2.266	2.330	2.197	2.278	2.358	2.261	2.042	1.936	(41)
(42)		Ebn at Noon		161	215	239	258	265	265	249	249	244	210	152	116	(42)
(43)		Edh at Noon		18	30	42	43	38	37	41	35	28	23	17	14	(43)
(44)	All-Sky Solar Radiation	RadAvg		197	506	987	1424	1559	1822	1654	1260	798	394	178	111	(44)
(45)		RadStd		19	37	62	108	161	192	140	119	110	40	36	15	(45)